Mixed ANOVA

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Introduction

A mixed ANOVA combines at least one between-subjects factor (treatment arm, sex) with at least one within-subjects factor (time point, condition). It is the standard analysis for longitudinal trials and for split-plot designs. The typical primary question is the **group-by-time interaction**: does the change over time differ between groups?

Prerequisites

Repeated-measures ANOVA, factorial designs.

Theory

For a design with between factor A (a levels, independent groups) and within factor B (b levels, measured on each subject):

$$Y_{ijk} = \mu + \alpha_i + \pi_{k(i)} + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk},$$

where $\pi_{k(i)}$ is the subject-level random effect nested in A.

Three effects are tested:

- Main effect of A (between): uses between-subjects MS as denominator.
- Main effect of B (within): uses within-subjects MS as denominator.
- A x B interaction: uses within-subjects MS as denominator.

Sphericity applies to the within factor (and to its interaction with A); Greenhouse-Geisser or Huynh-Feldt corrections are used when needed.

Assumptions

- Independent subjects within each between-group.
- Normal residuals, separately for within and between components.
- Homogeneous variance-covariance structure across between groups (Box's M, rarely formally tested).
- Sphericity of the within factor.

R Implementation

```r
library(afex); library(emmeans); library(ggplot2)
set.seed(2026)

# 30 subjects per arm, measured at 3 times
n <- 30
subj  <- rep(1:(2 * n), each = 3)
arm   <- factor(rep(rep(c("Placebo", "Active"), each = 3), n))
time  <- factor(rep(c("t0", "t1", "t2"), 2 * n))
```

```r
df <- data.frame(
  subject = factor(subj),
  arm, time,
  y = rnorm(6 * n, mean = 100, sd = 10) +
      ifelse(time == "t1", 3, 0) +
      ifelse(time == "t2", 5, 0) +
      ifelse(arm == "Active" & time == "t2", -8, 0)
)

fit <- aov_car(y ~ arm * time + Error(subject/time), data = df)
fit

# Interaction plot
emm <- emmeans(fit, ~ arm * time)
plot(emm, comparisons = TRUE)

# Simple effects of arm at each time
pairs(emm, by = "time", adjust = "bonferroni")
```

Output & Results

```
Anova Table (Type 3 tests)

 Effect          df   MSE      F     ges      p.value
 arm            1, 58   245   8.12   0.09    0.0062 **
 time           2, 116   45    9.55  0.03   <.001 ***
 arm:time       2, 116   45    4.72  0.02    0.011 *
```

Significant main effects of arm and time, and a significant interaction. Simple-effects analysis localises the interaction.

Interpretation

"A 2 (arm) x 3 (time) mixed ANOVA revealed a significant interaction, F(2, 116) = 4.72, p = 0.011, $\eta^2_G = 0.02$. Simple-effects comparisons showed that at time 2, the Active arm was 8.0 points lower than Placebo (p = 0.003)."

Practical Tips

- For longitudinal trials, the interaction is typically the primary endpoint.
- Sphericity corrections apply only to the within-subjects parts of the model.
- Missing measurements on a within-subject factor cause listwise deletion of that subject; prefer a linear mixed model for unbalanced data.
- Report the interaction plot; it is often more interpretable than the numeric ANOVA table.
- Simple main effects (one factor fixed at a level of the other) give the most interpretable decomposition when the interaction is significant.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**

- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests